

SIMATS ENGINEERING
SAVEETHA SCHOOL OF ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL – 1
PSEUDOCODE WRITING

Course Code	DSA0103	Course Title	Object Oriented Programming with C++
CO Assessed	CO1 (BL3 & BL4)	Assessment	Tool – 1
Weightage	40%	Total Marks	100
Student Name	Ranjitha. R	Reg No	192421285
Section / Batch	B.Tech - IT	Date	15-07-2026
CO Statement	Apply object-oriented programming principles and C++ constructs such as data types, operators, and control structures to design and write structured OOP-based programs. (BL3 & BL4)		

Total Questions: 10 Total Marks: 100

RUBRICS FOR EVALUATION

Criteria	Weightage	Level 4	Level 3	Level 2	Level 1
Problem Understanding	20%	All inputs correct	Minor missing	Partial	Incorrect
Algorithm Design	30%	Correct logic	Minor gaps	Partial	Wrong
Control Structures	20%	Efficient use	Minor issues	Basic use	Missing
Pseudocode Structure	10%	Well structured	Mostly clear	Hard to follow	Poor
Completeness	20%	All cases handled	Most cases	Basic	Incomplete

Code of Conduct: (192421285)

I Ranjitha - R (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Ranjitha

Code -

1

id -

absby abg . deab

Annexure A

QUESTIONS: 10

Total Marks: 100

PSUEDOCODE WRITING

1. A. A bank ATM checks whether withdrawal is possible based on balance. Fill in the blanks in the pseudocode to implement this logic correctly. (BL3)

```
START
READ balance, amount

IF amount ____ balance THEN
    PRINT "Transaction Approved"
    balance ← balance ____ amount
ELSE
    PRINT "Insufficient Balance"
END IF
```

```
PRINT balance
STOP
```

- B. An online store gives discount for high-value purchases. Fill in the blanks in the pseudocode to implement this logic correctly. (BL3)

```
START
READ amount

IF amount ____ 5000 THEN
    discount ← amount * 0.1
ELSE
    discount ← 0
END IF

final ← amount ____ discount
PRINT final
STOP
```

- C. A system checks eligibility based on attendance percentage. Check whether the student is eligible for writing exam? Complete the pseudocode by filling the blanks appropriately. (BL3)

```
START
READ attendance

IF attendance ____ 75 THEN
    PRINT "Eligible for Exam"
ELSE
    PRINT "Not Eligible"
```

Annexure A

QUESTIONS: 10

Total Marks: 100

PSUEDOCODE WRITING

1. A. A bank ATM checks whether withdrawal is possible based on balance. Fill in the blanks in the pseudocode to implement this logic correctly. (BL3)

```
START
READ balance, amount

IF amount ____ balance THEN
    PRINT "Transaction Approved"
    balance ← balance ____ amount
ELSE
    PRINT "Insufficient Balance"
END IF
```

```
PRINT balance
STOP
```

- B. An online store gives discount for high-value purchases. Fill in the blanks in the pseudocode to implement this logic correctly. (BL3)

```
START
READ amount

IF amount ____ 5000 THEN
    discount ← amount * 0.1
ELSE
    discount ← 0
END IF
```

```
final ← amount ____ discount
PRINT final
STOP
```

- C. A system checks eligibility based on attendance percentage. Check whether the student is eligible for writing exam? Complete the pseudocode by filling the blanks appropriately. (BL3)

```
START
READ attendance

IF attendance ____ 75 THEN
    PRINT "Eligible for Exam"
ELSE
    PRINT "Not Eligible"
```

```
END IF
STOP
```

D. A traffic system controls signals based on vehicle density. Depending on the vehicle density, calculate the overall traffic. Fill in the missing parts of the pseudocode to classify the temperature correctly. (BL3)

```
START
  READ vehicles

  IF vehicles ____ 50 THEN
    PRINT "Green Signal Extended"
  ELSE
    PRINT "Normal Timing"
  END IF
STOP
```

E. A company gives bonus based on years of experience. Based on experience fill the bonus. Complete the pseudocode by filling the blanks with suitable conditions. (BL3)

```
START
  READ years, salary
  IF years ____ 5 THEN
    bonus ← salary * 0.2
  ELSE
    bonus ← salary * 0.1
  END IF
  PRINT bonus
STOP
```

2. A smartphone needs to alert the user when the battery level becomes low. If the battery percentage is less than or equal to 20, the system should display "Low Battery"; otherwise, it should display "Battery OK". Write a pseudocode to read the battery level and display the appropriate message. (BL3)
3. A system validates user login credentials. If the entered username is "admin" and the password is "1234", the system should display "Login Successful"; otherwise, it should display "Access Denied". Write a pseudocode to implement this authentication process. (BL3)
4. A parking system calculates the parking fee based on the number of hours a vehicle is parked. If the parking duration is less than or equal to 2 hours, a fixed fee of ₹20 is charged; otherwise, the fee is calculated as ₹10 per hour. Write a pseudocode to read the number of hours and display the parking fee. (BL3)
5. A bank wants to verify ATM withdrawal requests. If the requested amount is less than or equal to the available balance, print "Transaction Approved"; otherwise print "Insufficient Balance". Develop the pseudocode for this process. (BL3)
6. A hospital assigns priority to patients based on their severity level. If the severity level is greater than or equal to 8, the patient is considered an "Emergency Case". If it is between 5 and 7, it is a "Moderate Case"; otherwise, it is a "Normal Case".

Case". Write a pseudocode to read the severity level and display the patient category. (BL3)

7. A vehicle monitoring system checks the fuel level and alerts the driver when fuel is low. If the fuel level is less than or equal to 5 liters, it should display "Refuel Needed"; otherwise, it should display "Sufficient Fuel". Write a pseudocode to read the fuel level and display the appropriate message. (BL3)
8. An online food delivery service calculates delivery charges based on order amount. If the order amount is ₹500 or more, delivery is free; otherwise, a delivery charge of ₹50 is added. Write a pseudocode to read the order amount and display the final payable amount. (BL3)
9. A company calculates overtime pay for employees. If an employee works more than 40 hours per week, they are paid ₹200 per extra hour; otherwise, no overtime is given. Write a pseudocode to read total working hours and calculate the overtime pay. (BL3)
10. A system evaluates password strength. If the password length is less than 6 characters, it is "Weak". If it is between 6 and 10 characters, it is "Moderate". If it is more than 10 characters, it is "Strong". Write a pseudocode to read the password length and display its strength. (BL3)

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear

PSEUDO CODE:

1) A) ATM WITHDRAWAL:

START

READ balance, amount

IF amount $\leq$ balance THEN

PRINT "Transaction Approved"

balance $\leftarrow$ balance - amount

ELSE

PRINT "Insufficient Balance"

END IF

PRINT balance

STOP

B) ONLINE STORE DISCOUNT:

START

READ amount

IF amount $\geq$ 5000 THEN

discount $\leftarrow$ amount $\times$ 0.1

ELSE

discount $\leftarrow$ 0

END IF

final $\leftarrow$ amount - discount

PRINT final

STOP

PSEUDO CODE:

1) A) ATM WITHDRAWAL:

```
START
READ balance, amount
IF amount  $\leq$  balance THEN
    PRINT "Transaction Approved"
    balance  $\leftarrow$  balance - amount
ELSE
    PRINT "Insufficient Balance"
END IF
PRINT balance
STOP
```

B) ONLINE STORE DISCOUNT:

```
START
READ amount
IF amount  $\geq$  5000 THEN
    discount  $\leftarrow$  amount * 0.1
ELSE
    discount  $\leftarrow$  0
END IF
final  $\leftarrow$  amount - discount
PRINT final
STOP
```

Code -

1. id2 + id1,

absby aba . aab

C) ATTENDANCE ELIGIBILITY

```
START
READ attendance
IF attendance >= 75 THEN
    PRINT "Eligible for Exam"
ELSE
    PRINT "Not Eligible".
END IF
STOP
```

D) TRAFFIC SIGNAL

```
START
READ vehicles
IF vehicles > 50 THEN
    PRINT "Green Signal Extended"
ELSE
    PRINT "Normal Timing"
END IF
STOP
```

E) EMPLOYEE BONUS

```
START
READ years, salary
IF years >= 5 THEN
    bonus ← salary * 0.2
```


C) ATTENDANCE ELIGIBILITY

START

READ attendance

IF attendance $\geq$ 75 THEN

PRINT "Eligible for Exam"

ELSE

PRINT "Not Eligible".

END IF

STOP

D) TRAFFIC SIGNAL

START

READ vehicles

IF vehicles $>$ 50 THEN

PRINT "Green Signal Extended"

ELSE

PRINT "Normal Timing"

END IF

STOP

E) EMPLOYEE BONUS

START

READ years, salary

IF years $\geq$ 5 THEN

bonus $\leftarrow$ salary * 0.2

2) LOW BATTERY ALERT

START

READ Battery

IF battery < 20 THEN

PRINT "Low Battery"

ELSE

PRINT "Battery OK"

END IF

STOP

3) LOGIN AUTHENTICATION

START

READ username, password

IF username = "admin" AND password = "1234" THEN

PRINT "Login Successful"

ELSE

PRINT "Access Denied"

END IF

STOP

4) PARKING FEE

START

READ hours

IF hours < 2 THEN

fee $\leftarrow 20$

ELSE

fee $\leftarrow$ hours * 10

END IF

PRINT fee

STOP

5) ATM WITHDRAWAL VERIFICATION:

START

READ balance, amount

IF amount $\leq$ balance THEN

PRINT "Transaction Approved"

ELSE

PRINT "Insufficient Balance"

END IF

STOP

6) HOSPITAL PATIENT PRIORITY

START

READ Severity

IF Severity $\geq$ 8 THEN

PRINT "Emergency case"

ELSE

PRINT "Normal case"

END IF

STOP

7) FUEL LEVEL CHECK

START

READ fuel

IF fuel < 5 THEN

PRINT "Refund Needed"

ELSE

PRINT "Sufficient Fuel"

END IF

STOP

8) ONLINE FOOD DELIVERY CHARGES:

START

READ amount

IF amount ≥ 500 THEN

final $\leftarrow$ amount

ELSE

final $\leftarrow$ amount + 50

END IF

PRINT final

STOP

9) EMPLOYEE OVERTIME PAY

START

READ hours


```
IF hours > 40 THEN
    overtime ← (hours - 40) * 200
ELSE
    overtime ← 0
END IF
PRINT overtime
STOP
```

10) PASSWORD STRENGTH:

```
START
READ length
IF length < 6 THEN
    PRINT "Weak"
ELSE IF length ≤ 10 THEN
    PRINT "Moderate"
ELSE
    PRINT "Strong"
END IF
STOP
```